

Directions: For the following situations, match the distribution type with the corresponding situation. Each distribution will be used only once.

- A. Normal Probability Distribution
- B. Combining Normal Random Variables Distribution
- C. Sampling Distribution for Sample Proportions
- D. Sampling Distribution for a Difference in Sample Proportions
- E. Sampling Distribution for Sample Means
- F. Sampling Distribution for a Difference in Sample Means

_____ 1) A researcher is comparing the effectiveness of two different medications in reducing blood pressure. They randomly assign 50 patients to receive Medication A and 50 patients to receive Medication B. What is the probability that the difference in the mean reduction in blood pressure between the two groups is less than 5 mmHg?

_____ 2) In a factory, the weights of products produced follow a normal distribution with a mean of 500 grams and a standard deviation of 20 grams. If a package contains 10 of these products, what is the probability that the total weight of the package is less than 4950 grams?

_____ 3) In a study on the heights of sunflowers, it's found that the heights follow a normal distribution with a mean of 150 centimeters and a standard deviation of 20 centimeters. A researcher randomly selects a sample of 25 sunflowers from a field. What is the probability that the average height of the sample is greater than 155 centimeters?

_____ 4) A company manufactures light bulbs with a mean lifespan of 800 hours and a standard deviation of 50 hours. What is the probability that a randomly selected light bulb will last more than 850 hours.

_____ 5) In a study comparing the effectiveness of two different advertising strategies, Strategy A and Strategy B, a marketing team wants to determine if there's a significant difference in the proportion of customers who make a purchase after seeing the ad. In a sample of 200 customers exposed to Strategy A, 40 customers made a purchase. In a sample of 250 customers exposed to Strategy B, 65 customers made a purchase. What is the probability that the difference in the proportion of customers who made a purchase between the two strategies is greater than 0.05?

_____ 6) A survey was conducted to see the proportion of smartphone users who prefer iPhone. A research team randomly selects 800 smartphone users. Among them, 520 users prefer iPhone. What is the probability that in another random sample of 800 smartphone users, the proportion preferring iPhone is less than 0.6?

7) In a clinical trial investigating the effectiveness of a new medication for allergies in dogs, 60 dogs were randomly assigned to two groups: 30 dogs went to Group M, where they received the new medication, while 30 dogs went to Group P, where they received a placebo. After a month of treatment, the severity of allergy symptoms in each dog was assessed using a standardized scale. The mean score for Group M was 3.5 units with a standard deviation of 1.2 units, while the mean score for Group P was 4.2 units with a standard deviation of 1.5 units. A higher score means that more allergy symptoms were observed. What is the probability that the difference in the mean scores between the two groups (Group M – Group P) is greater than 0?